

StochLean

Markov Processes, Kernels, Semigroups & Chain Convergence

Work Handoff Design - Version 0.1

Primary mathematical coverage: Klenke 3rd ed. (2020), Ch.14.4, Ch.17-18, and Feller/strong-Markov interfaces in Ch.21.
Implementation organization is functional. Klenke chapter numbers are provenance only.

NORMATIVE PACKAGE ORDER. Build generic information flow and TransitionSystem first, then homogeneous MarkovSemigroup and Markov semantic bridges, then countable-chain recurrence/invariance/periodicity/coupling, then bounded Q-matrix/uniformization, and finally Feller interfaces. No later phase may create temporary duplicates of an unfinished earlier owner.

1. Purpose, Package Boundary, and Status Vocabulary

- This document is the implementation contract for the StochLean Markov package. It consolidates all approved design decisions from the Markov design rounds and is normative for Work.
- Processes remain raw functions $X : T \rightarrow \Omega \rightarrow E$. Kernels remain Mathlib ProbabilityTheory.Kernel. No StochasticProcess, MarkovProcess, TransitionKernel, or transition-matrix wrapper structure is introduced.
- General two-parameter evolution is named TransitionSystem. One-parameter time-homogeneous evolution is named MarkovSemigroup.
- The package consumes: ProcessCore filtration/stopping/path semantics; Ch01-08 condExp/condDistrib/Kernel/tail/Poisson; DiscreteMartingale convergence where needed; MeasureConvergence/ProjectiveConstruction KET and weak-convergence infrastructure.
- MCMC, finite-state spectral rates, general Banach generator theory, full Skorokhod D topology, and unrelated ergodic theory are not current core blockers unless an audited proof of a current theorem genuinely requires a canonical prerequisite.

Status	Meaning / Work action
REUSE	Exact canonical functionality exists. Map exact declaration/import; never mirror locally.
BRIDGE / REUSE-BRIDGE	Core exists; add only a thin reusable theorem/adaptor for a real API gap.
BUILD	Confirmed foundation gap; implement the mathematically correct statement.
AUDIT-FIRST	Search pinned Mathlib -> complete local tree -> approved external ledger before coding.
EXTERNAL-PROOF-AUDIT	The textbook points to an external proof. Audit exact source and assumptions before formalization.
DEFER	Record owner/reason/unlock condition; no current proof attempt.
APPLICATION	Coverage/acceptance example; not a foundation blocker.

2. Global Non-Negotiable Rules

- No-duplicate gate: before every public declaration search (1) the active pinned Mathlib revision, (2) the entire local StochLean tree including concurrently owned packages, then (3) the approved-external ledger. A usable semantic equivalent forbids a parallel definition/theorem.
- Concurrent-package rule: an unfinished declaration assigned to another canonical owner is a blocker, not permission to create a temporary copy.
- No proof holes or semantic cheating: no sorry/admit/project axioms, vacuous assumptions, conclusion-shaped hypotheses, surrogate models, or exploitation of totalized fallback values.
- Semantics: preserve a.e. vs pointwise, bounded vs a.s.-finite stopping times, Markov semantic property vs chosen transition representation, f.d.d./coordinate-product/path-space law distinctions, and finite vs extended values.

- External trust boundary: exact source, edition/chapter/theorem, exact code revision if Lean code, license, compile status, axiom/dependency closure, and audited/un-audited status must be recorded.
- Canonical-owner rule: generic probability utilities such as coupling and future sigma-fields live in generic modules, not Markov-private duplicates.
- Source-correction rule: the correction table in this handoff overrides literal OCR/printed text. Work must implement the corrected mathematics.

3. Mandatory Proof-Source Protocol for Work

FIXED POLICY FOR THIS AND FUTURE PACKAGES. When a Klenke result is proved only by an external citation, Work must first retrieve the cited proof or another explicit public proof with exact bibliographic locator. If the concrete proof text cannot be accessed and audited, mark the item DEFER and skip it. Do not spend autonomous effort re-proving an externally cited textbook theorem merely to close the package.

- For every BUILD/EXTERNAL-PROOF-AUDIT row, the handoff/report should state a proof strategy and the preferred source route. If an external proof is available, audit it first and formalize that proof (or the minimal dependency closure) into the canonical owner.
- If the external source is accessible but its statement/hypotheses differ, report the mismatch before coding. Do not silently strengthen/weaken Klenke.
- If the source is inaccessible in concrete form, status becomes DEFER with source locator, reason, future owner, and downstream impact. Work must not substitute an original proof attempt.
- For results fully proved in Klenke, the textbook proof can be formalized or replaced by existing canonical library results. The no-duplicate gate still takes precedence.

External theorem / proof target	Primary lookup route for Work	Action if accessible	If inaccessible
Feller pointwise definition $\Leftrightarrow$ C0 sup-norm strong continuity	D. Revuz & M. Yor, Continuous Martingales and Brownian Motion, 3rd ed. (1999), Ch. III, Proposition 2.4 is a standard cited locator. Work must verify the exact statement in the source before use.	Audit hypotheses; formalize equivalence as theorem, keeping Klenke Def.21.26 as canonical predicate.	DEFER the equivalence theorem; do not invent a proof.
RCLL/cadlag Feller realization used before Klenke Thm.21.27	L.C.G. Rogers & D. Williams, Diffusions, Markov Processes and Martingales, Vol.1: Foundations, 2nd ed./2000 reprint, Ch. III.7ff; Klenke ref. [147].	Audit exact construction and minimal prerequisites; formalize in canonical path/filtration owners.	DEFER the realization; record blocker.
Strong Markov property / Klenke Thm.21.27	Rogers-Williams Vol.1, Ch. III.8ff; or Revuz-Yor, Ch. III, Theorem 2.7; Klenke refs. [147], [145].	Audit exact theorem and hypotheses, then formalize to close 21.27.	DEFER 21.27; do not attempt an independent proof.
Any future externally cited theorem	Use the exact citation in Klenke first; if Klenke is imprecise, locate a public standard source with precise theorem/section.	Audit $\rightarrow$ formalize/import according to canonical ownership.	DEFER + maintenance ledger entry.

4. Information Flow and General TransitionSystem

4.1 Future sigma-field is a mandatory generic deliverable

- Create/bridge futureMeasurableSpace X_t with mathematical meaning $\sigma(X_s : s \geq t)$. Place it in generic Process/Filtration ownership, not Markov-private code.
- Required API: $\sigma(X_t) \leq \text{futureMeasurableSpace } X_t$; $t \leq u$ implies $\text{futureMeasurableSpace } X_u \leq \text{futureMeasurableSpace } X_t$; minimality against any measurable space making all $X_s, s \geq t$ measurable.
- For discrete time, bridge the canonical tail sigma-field to $\text{intersection_n futureMeasurableSpace } X_n$. Do not redefine tail sigma-field.

4.2 TransitionSystem first, MarkovSemigroup second

- The general inhomogeneous object is a two-parameter forward family $K_{\{r,s\}}$; public naming: TransitionSystem / IsMarkovTransitionSystem.
- Avoid totalized values for $r > s$. Preferred Lean representation: $K : (r s : T) \rightarrow r \leq s \rightarrow \text{Kernel } E \ E$ (or an equivalent proof-indexed forward-pair representation).

- Core assumptions need only a suitable preorder for composable triples. For $r \leq s \leq t$: $K_{\{r,t\}} = K_{\{s,t\}} \circ_k K_{\{r,s\}}$; $K_{\{t,t\}} = \text{Kernel.id}$; every $K_{\{r,s\}}$ is Markov.
- The diagonal identity is a canonical completion of Klenke Def.14.42 and must be recorded as such.

KERNEL ORIENTATION - WORK MUST REPORT THIS EXPLICITLY. Mathlib defines $\eta \circ_k \kappa$ as: first κ , then η . Traditional probability multiplication $K_{\{r,s\}} K_{\{s,t\}}$ means first $K_{\{r,s\}}$, then $K_{\{s,t\}}$. Hence the Lean Chapman-Kolmogorov law is $K_{\{r,t\}} = K_{\{s,t\}} \circ_k K_{\{r,s\}}$. StochLean uses Mathlib $\circ_k$ and does NOT define a second multiplication operator.

Probability notation / chronology	Mathlib / StochLean Lean meaning
$K_{\{r,s\}} K_{\{s,t\}}$	$K_{\{s,t\}} \circ_k K_{\{r,s\}}$
first $K_{\{r,s\}}$, then $K_{\{s,t\}}$	$K_{\{s,t\}} \circ_k K_{\{r,s\}}$
$K_{\{r,s\}} K_{\{s,t\}} = K_{\{r,t\}}$	$K_{\{r,t\}} = K_{\{s,t\}} \circ_k K_{\{r,s\}}$
homogeneous $K_s K_t = K_{\{s+t\}}$	$\kappa(s+t) = \kappa t \circ_k \kappa s$
operator action on functions	$P_{\{s+t\}} = P_s \circ P_t$

5. Markov Semantic Property, Homogeneous MarkovSemigroup, and Construction

5.1 HasMarkovProperty is kernel-free semantic data

- Canonical predicate: $\text{HasMarkovProperty } X \text{ F P}$. Include adaptedness and event-form conditional-expectation equality: for $s \leq t$ and measurable A , $E[1_{\{X_t \text{ in } A\}} | \mathcal{F}_s] = E[1_{\{X_t \text{ in } A\}} | \sigma(X_s)]$ a.e.
- Event form is the definition; bounded-measurable test-function form is a load-bearing theorem via monotone-class arguments.
- Past/future conditional independence is a characterization theorem, using Mathlib $\text{CondIndep}/i\text{CondIndep}$. For natural filtration: past σ -field and future $\text{MeasurableSpace } X \text{ t}$ are conditionally independent given $\sigma(X_t)$.
- Do not require the ambient sample space to be StandardBorel merely to state Markov property.

5.2 MarkovSemigroup

- One-parameter time-homogeneous family name: $\text{MarkovSemigroup} / \text{IsMarkovSemigroup}$. It is a property on $\kappa : T \rightarrow \text{Kernel } E$, not a new kernel object.
- Core time algebra: $\text{AddCommMonoid } T$ unless a downstream theorem needs order/topology. Laws: $\kappa 0 = \text{Kernel.id}$, every κt Markov, and $\kappa(s+t) = \kappa t \circ_k \kappa s$.
- Derive the opposite-order equality and commuting kernels from time commutativity. Discrete-time acceptance: $n \mapsto \kappa^n$ must instantiate the semigroup using Mathlib Kernel.pow_add .
- Do not force a MonoidHom representation into the public core. It obscures chronology because Mathlib composition order is opposite traditional probability multiplication.

5.3 Transition representation and existence

- $\text{HasTransitionSystem } X \text{ F P K}$ expresses $E[1_{\{X_s \text{ in } A\}} | \mathcal{F}_r] = K_{\{r,s\}}(X_r, A)$ a.e. for $r \leq s$. This implies HasMarkovProperty .
- Homogeneous $\text{HasTransitionSemigroup}$ is a specialization, not the definition of Markov property.
- Construct canonical Markov processes from consistent finite laws / TransitionSystem using the previous StandardBorel KET facade. The Markov package must not implement a second projective extension.
- For discrete one-step chains, build/bridge $\text{markovChainPathKernel } \kappa : \text{Kernel } E (\text{Nat} \rightarrow E)$ as a thin specialization of Mathlib Ionescu-Tulcea. No MarkovChain structure.

6. Countable-State Chain Core: Hitting, Recurrence, Invariance, Periodicity, Coupling

6.1 Reachability and irreducibility

- Mathlib $\text{Kernel.IsIrreducible Measure.count } \kappa$ is canonical. Do not define $\text{IsIrreducibleMarkovChain}$.
- If absent, add thin $\text{Kernel.CanReach } \kappa x y := \exists n, (\kappa^n x) \{y\} > 0$. Prove $\text{IsIrreducible count } \kappa$ iff pairwise CanReach under measurable-singleton hypotheses.
- Klenke return probability $F(x,y) > 0$ uses strictly positive time and is NOT definitionally the same as CanReach on $x=y$ because CanReach permits $n=0$. The diagonal positive-return proof is mandatory.
- Klenke weak irreducibility is a distinct countable-chain notion; never alias it to $\text{Kernel.IsIrreducible}$.

6.2 Strict-positive hitting and recurrence

- Reuse Mathlib hittingAfter. For discrete time, firstPositiveHittingTime X A is definitionally/thinly hittingAfter X A 1. Keep WithTop Nat; never use untop/default on the infinite event as mathematics.
- Return probability $F_{\kappa}(x,y) = P_x(\tau_y^+ < \infty)$. Mean return time $m_{\kappa}(x)$ is an ENNReal lintegral of the ENat/WithTop-Nat return time under the canonical chain path law.
- State predicates: RecurrentAt iff $F(x,x)=1$; PositiveRecurrentAt iff $m(x)<\infty$; NullRecurrentAt = recurrent and not positive recurrent; TransientAt iff $F(x,x)<1$.
- Prove tail-sum identity $m(x)=\sum_n P(\tau_x^+>n)$ and positive recurrence $\rightarrow$ recurrence.

6.3 Green function and recurrence propagation

- Kernel.green $\kappa x y := \sum_n (\kappa^n x) \{y\}$ in ENNReal if no canonical API exists.
- Load-bearing theorems: Green = expected number of visits; recurrent at x iff $G(x,x)=\infty$; recurrence and positive recurrence propagate along communication as in Klenke Ch.17.

6.4 Excursion occupation measure and Kac formula

- Canonical construction: excursionOccupationMeasure $\kappa x = \sum_n (X_n)_* (P_x^{\kappa} \text{ restricted to } \{\tau_x^+>n\})$. This is a measure-series construction, not singleton-weight assembly.
- Required: occupation interpretation; total mass equals mean return time; mass of singleton $\{x\}$ equals 1; under recurrence, Kernel.Invariant $\kappa \mu_x$.
- When x is positive recurrent, normalize μ_x by its finite mass to obtain an invariant probability. Under irreducibility, prove uniqueness and Kac formula $\pi(\{x\}) = 1 / E_x[\tau_x^+]$.

6.5 Periodicity

- Define transitionTimes $\kappa x y = \{n : (\kappa^n x)\{y\}>0\}$; period(x)=gcd transitionTimes(x,x); aperiodic iff period=1.
- Load-bearing: irreducible states share period; eventual positivity on admissible residue classes (Klenke Lemma 18.3 level), not merely equality of periods. Cyclic decomposition can follow.

7. Coupling, Source Corrections, and TV Convergence

7.1 Generic coupling ownership

- If absent, generic Probability/Coupling/Basic owns IsCoupling $\gamma \mu \nu$: the two coordinate marginals of γ are μ and ν . Product measure is a witness. Do not define coupling only inside Markov.
- Path coupling for a Markov chain is simply a coupling of the canonical path laws P_x^{κ} and P_y^{κ} ; no bivariate-process wrapper structure is required.
- Successful path coupling follows Klenke Def.18.5 tail-mismatch criterion; prove equivalence with eventual equality almost surely. Pair-level and global existence may be separated.

7.2 Independent coalescent - hard semantic source safeguard

DO NOT COPY THE OCR/PIECEWISE FORMULA. Define the independent coalescent from its semantics. If current states $x \neq y$, the next pair law is the product kernel $\kappa(x,du) \times \kappa(y,dv)$. If current states $x = y$, push $\kappa(x,dz)$ through $z \mapsto (z,z)$, so the diagonal is absorbing. Only after this semantic definition may Work derive the singleton piecewise transition formula. Any OCR condition that loses a $\neq$ sign is ignored.

- Recommended object: independentCoalescentKernel κ on $E \times E$, then derive its canonical pair-chain path law and push to a coupling of two E^{Nat} path laws.
- Coalescence time is the ordinary hitting time of the diagonal from time 0. For this kernel, successful coupling is equivalent to a.s. finite diagonal hitting.

7.3 Klenke (17.30) factor-2 correction - hard source safeguard

SOURCE CORRECTION. Klenke (17.29) uses the full signed-measure TV norm $\sup_{\|f\|_{\infty} \leq 1} \int f d(P-Q)$. Under that convention, the printed coupling identity (17.30) is missing a factor 2. Work must NOT copy it literally. Correct: $\|P-Q\|_{TV} = 2 \inf_{\gamma \in K(P,Q)} \gamma\{x \neq y\}$. The load-bearing inequality is $\|P-Q\|_{TV} \leq 2 \gamma\{x \neq y\}$. Klenke Thm.18.12 itself uses the factor-2 inequality, confirming the normalization.

- Use Mathlib signed-measure total variation/variation as the canonical norm. Do not introduce an ambiguous TVDistance with a hidden half-normalization.

- Corrected maximal-coupling equality is a generic BUILD-CANDIDATE; the factor-2 inequality is mandatory/load-bearing for 18.12.

7.4 Final countable convergence theorem

- Prove the Klenke chain: irreducible + positive recurrent + aperiodic $\rightarrow$ independent coalescent successful $\rightarrow$ TV convergence.
- The final target is full Klenke Thm.18.13 equivalence between aperiodicity and TV convergence from every state / some state / every initial probability. Do not weaken the target to weak convergence; weak convergence is only a corollary.

8. Continuous-Time Countable Chains: Q-Matrix and Uniformization

8.1 IsQMatrix - countability policy fixed

- Fixed choice: IsQMatrix q itself is summability-based and does NOT require [Countable E]. Countability/discrete measurability is imposed only on uniformization and countable-chain construction theorems that actually need it.
- $q : E \rightarrow E \rightarrow \text{Real}$. Conditions: $q(x,y) \geq 0$ for $x \neq y$; off-diagonal row is summable; $q(x,x) = -\sum_{y \neq x} q(x,y)$. Prefer a subtype $y \neq x$ for the sum.

Q-MATRIX OCR SAFEGUARD. Any extracted text suggesting $q(x,y) \geq 0$ for $x=y$ is an OCR loss of $\neq$. Positivity is off-diagonal. The diagonal is nonpositive and equals minus the off-diagonal row sum.

- Define/bridge $\text{exitRate} = -q(x,x)$. Bounded exit rates are represented by an explicit dominating Lambda , not a foundational sSup .

8.2 Uniformization

- If $\text{Lambda} > 0$ and dominates all exit rates, define $P_{\text{Lambda}}(x,y) = q(x,y)/\text{Lambda}$ off-diagonal and $1 + q(x,x)/\text{Lambda}$ on the diagonal; produce a Markov kernel.
- Mandatory $\text{Lambda} = 0$ branch: domination implies $q = 0$; use Kernel.id . Never rely on division-by-zero totalization.
- Uniformized semigroup: $K_t = \exp(-\text{Lambda } t) \sum_n (\text{Lambda } t)^n / n! P_{\text{Lambda}}^n$, equivalently a Poisson mixture of kernel powers. Reuse the existing Poisson package and kernel-sum machinery.
- Prove IsMarkovSemigroup with the StochLean/Mathlib orientation $K_{\{s+t\}} = K_t \circ K_s$.

8.3 HasQMatrix and Klenke Thm.17.25

- $\text{HasQMatrix } \kappa q$ is a relation: right derivative at $t=0$ of singleton transition probabilities, converted safely to Real, equals $q(x,y)$ for all x,y , including the diagonal.
- Prove the uniformized semigroup recovers q . Bounded-Q existence is BUILD. Uniqueness is also mandatory: Klenke says unique Markov process; in StochLean formalize unique transition semigroup + unique canonical process law, not raw equality of processes on unrelated sample spaces.
- Preferred uniqueness strategy: derive the Kolmogorov integral/differential equation from semigroup + right derivative and show the unique bounded solution is the uniformization series. Do not merely assert $\exp(tQ)$.
- Poisson process Q-matrix is an acceptance test and must bridge to the earlier canonical Poisson transition law.

9. Feller Semigroups and Strong-Markov Realization

9.1 Canonical definition

- Use Mathlib $C_0(E, \mathbb{R}) = \text{ZeroAtInftyContinuousMap}$. No local $C_0/\text{FellerFunction}$ type.
- Canonical IsFellerSemigroup follows Klenke Def.21.26: it includes IsMarkovSemigroup ; κ_t maps every f in $C_0(E)$ back to $C_0(E)$; and $P_t f(x) \rightarrow f(x)$ pointwise as $t \downarrow 0$.
- Do not replace the definition by sup-norm strong continuity. The strong-continuity formulation is to be proved equivalent as a theorem.

9.2 Mandatory equivalence with C_0 sup-norm strong continuity

- Formalize or reuse the theorem that Klenke-style C_0 preservation + pointwise continuity at zero, together with the Markov contraction semigroup structure, implies $\|P_t f - f\|_{\infty} \rightarrow 0$; the converse is immediate.
- Proof-source priority: first audit Revuz-Yor, Ch.III, Proposition 2.4 (standard cited locator; verify exact edition/statement). If the proof text is accessible and matches, formalize it or a minimal equivalent proof. If inaccessible, mark DEFER and do not attempt an original proof.
- Once preservation is available, define/bridge the induced C_0 operator. Mandatory: contraction $\|P_t f\| \leq \|f\|$ and operator semigroup $P_{\{s+t\}} = P_s \circ P_t$; this order follows from the kernel orientation.

- Use the previous measure-convergence package to derive fixed-time weak continuity $x_n \rightarrow x \Rightarrow \kappa_t(x_n, \cdot) \Rightarrow \kappa_t(x, \cdot)$, and stochastic continuity $\kappa_t(x, \cdot) \Rightarrow \delta_x$ as $t \downarrow 0$. Do not re-prove Portmanteau locally.

9.3 Klenke Theorem 21.27: external-proof audit -> formalization if accessible

- Klenke explicitly points to Rogers-Williams Vol.1, Ch.III.7ff for the RCLL realization step; and to Rogers-Williams Vol.1, Ch.III.8ff or Revuz-Yor Ch.III, Theorem 2.7 for strong Markov. These are the primary Work lookup routes.
- If the cited proof(s) are concretely accessible, audit exact hypotheses and dependency closure. If correct, formalize the missing canonical prerequisites and then prove the StochLean theorem: a Feller semigroup on a locally compact Polish space admits an RCLL/cadlag strong-Markov realization with the prescribed transition kernels.
- If the proof text cannot be accessed, mark the item DEFER with locator and blocker information and move on. Work must not independently reconstruct 21.27.
- The theorem is existence/version semantics. Do not claim every realization of the same semigroup is automatically RCLL.
- Any required usual augmentation, Doob regularization, cadlag modification, or random-time measurability theorem must be placed in its canonical generic owner. Do not hide Markov-private duplicates in Feller.lean.

10. Consolidated Source/OCR Corrections and Semantic Warnings

Issue	Normative Work action
Kernel multiplication vs Mathlib composition	Traditional $K_{\{r,s\}}K_{\{s,t\}}$ translates to $K s t \circ _k K r s$. Report the convention prominently; implement only Mathlib $\circ _k$.
Klenke (17.30) TV coupling identity	Insert factor 2 under Klenke (17.29) full-TV norm. Never copy printed formula literally.
Klenke Example 18.6 independent-coalescent piecewise OCR	Ignore potentially corrupted cases. Define semantically: product off diagonal; diagonal pushforward when equal; derive formula afterward.
Klenke Q-matrix extracted $x=y / x!=y$	Off-diagonal $q_{xy} \geq 0$ and row sum over $y! = x$. Diagonal is negative exit rate.
WithTop stopping-time evaluation	Do not use untop/default fallback on $\tau = \infty$ as a mathematical value.
Klenke 21.27 proof boundary	External proof must be retrieved and audited before formalization. If inaccessible -> DEFER, no autonomous reconstruction.
Feller definitions	Keep Klenke pointwise C0 definition canonical; prove strong-continuity equivalence separately, never conflate by documentation only.

11. Functional Ownership and Candidate Module Boundary

Probability/Process/Filtration/Future.lean # mandatory generic future sigma-field
 Probability/Markov/Basic.lean
 Probability/Markov/TransitionSystem.lean # general two-parameter first
 Probability/Markov/Semigroup.lean # homogeneous MarkovSemigroup
 Probability/Markov/Transition.lean # semantic <-> kernel relations
 Probability/Markov/StrongMarkov.lean # countable-time theorem + audited Feller interface
 Probability/Markov/Chain/PathLaw.lean # thin Ionescu-Tulcea specialization
 Probability/Markov/Countable/{Hitting,Recurrence,Green,Excursion,Invariant,Periodicity,Coalescent,Convergence}.lean
 Probability/Markov/Generator/{QMatrix,Uniformization}.lean
 Probability/Markov/Feller.lean
 Probability/Coupling/{Basic,TotalVariation}.lean # generic, only if real gap survives audit

NOT A FILE CREATION CHECKLIST. Every candidate file disappears if the pinned Mathlib/local/external audit already provides the canonical functionality. Do not create local mirrors of Kernel, Invariant, IsReversible, IsIrreducible, hitting-time, conditional-independence, KET, Portmanteau, or C0 infrastructure.

12. Fixed Blocker & Deferred-Dependency Ledger - Maintenance Format

FIXED FORMAT FOR THIS AND ALL FUTURE PACKAGE HANDOFFS. Every Work report must contain a blocker/deferred ledger with one row per unresolved or intentionally deferred dependency. A vague TODO list is not acceptable.

ID	Content	Owner	Status	Why blocked/deferred	Unlock condition	Downstream impact
MK-B01	Arbitrary-index StandardBorel KET external audit if inherited package not closed	Measure/Projective package	INHERITED	Previous package external audit not complete	Previous package G5 passes	Canonical Markov process construction
MK-B02	futureMeasurableSpace generic API	Process/Filtration	CURRENT	Required by conditional-independence characterization	Build/bridge + tests	Markov semantic API
MK-B03	bounded Q-matrix -> unique semigroup	Markov/Generator	CURRENT	Uniqueness proof is load-bearing	Uniformization existence + uniqueness proof	Continuous-time countable chains
MK-B04	Klenke-Feller -> C0 strong continuity	Markov/Feller or generic semigroup bridge	EXTERNAL-PROOF-AUDIT	Proof source must be audited	Accessible Revuz-Yor/equivalent proof audited + formalized	Feller operator/generator compatibility
MK-B05	Feller RCLL realization	Path/Filtration + Markov/Feller	EXTERNAL-PROOF-AUDIT	Klenke refers outward	Rogers-Williams III.7ff proof accessible/audited	Klenke 21.27
MK-B06	Feller strong Markov theorem	Markov/StrongMarkov	EXTERNAL-PROOF-AUDIT	Klenke refers outward	Rogers-Williams III.8ff or Revuz-Yor III.2.7 accessible/audited	Klenke 21.27
MK-D01	General Banach generator theory	Future semigroup/SDE layer	DEFERRED	Not needed for current Q-matrix/Feller core	Future generator package	SDE/martingale problem
MK-D02	Full Skorokhod D topology	Functional weak convergence package	DEFERRED	Path-space topology is separately owned	Future path-space package	Donsker/process limits
MK-D03	MCMC applications	Application package	DEFERRED	Not foundation blocker	After invariant/convergence APIs stable	Sampling applications
MK-D04	Finite-state spectral convergence rates	Finite Markov spectral extension	DEFERRED	Not needed for TV convergence theorem	Future finite-state spectral work	Rates only

- Required final row status vocabulary: RESOLVED / STILL-BLOCKED / DEFERRED (owner + reason). Work must update every row before milestone completion.
- For an external-proof row: if source text is inaccessible, change to DEFERRED immediately and continue other tasks. Do not convert it to an autonomous proof project.

13. Normative Work Implementation Order

Phase	Required deliverable
A	Full duplicate/API audit: pinned Mathlib -> complete local StochLean -> approved external ledger. Populate exact declaration map and blocker ledger.
B	Build/bridge futureMeasurableSpace; close decreasing-family and tail bridges.
C	Implement TransitionSystem general two-parameter API and orientation tests; then MarkovSemigroup homogeneous specialization.
D	Implement HasMarkovProperty event semantics, bounded-function theorem, conditional-independence characterization, transition relations, and canonical process/path-law construction through KET/Ionescu-Tulcea.
E	Countable chain: reachability/irreducibility bridge; strict-positive hitting; return probabilities; mean return; Green; recurrence propagation.
F	Excursion occupation measure -> invariant measure -> positive recurrence equivalence -> uniqueness -> Kac formula.
G	Periodicity/eventual positivity/cyclic decomposition needed by product-chain irreducibility.
H	Generic coupling + corrected TV normalization; semantic independent-coalescent construction; successful coupling; Klenke 18.11/18.12/full 18.13.
I	Q-matrix summability API; explicit uniformization rate; Lambda=0 branch; Poisson-mixture semigroup; HasQMatrix; bounded-Q existence and uniqueness; Poisson acceptance.
J	Feller predicate on Mathlib C0; contraction/operator laws; weak-continuity bridges; external proof audit for strong-continuity equivalence and Klenke 21.27 following Section 3 policy.
K	Final source/correction audit; blocker ledger update; no-sorry/axiom/dependency inspection; semantic CI suite; minimal imports; longevity/upstream report.

14. Mandatory Semantic Regression and Acceptance Suite

1. futureMeasurableSpace is decreasing in time; its discrete intersection recovers the existing tail sigma-field.
2. TransitionSystem only has forward-time kernels; no arbitrary $r > s$ totalization is used.
3. Kernel orientation: test a noncommuting two-step example showing $K_{\{r,t\}} = K_{\{s,t\}} \circ K_{\{r,s\}}$, not the reverse.
4. HasMarkovProperty is not defined by existence of a chosen kernel; event and bounded-function forms agree.
5. Past/future conditional independence uses the mandatory future sigma-field and Mathlib CondIndep.
6. Discrete κ^n instantiates MarkovSemigroup and matches Mathlib Chapman-Kolmogorov.
7. CanReach permits $n=0$; Klenke $F(x,x) > 0$ requires a positive-time return and must be proved separately.
8. firstPositiveHittingTime uses hittingAfter ... 1 and preserves infinity; no untop fallback.
9. mean return time is extended-valued; positive recurrent $\rightarrow$ recurrent.
10. excursion occupation measure has mass $m(x)$, singleton mass 1, and is invariant under recurrence.
11. irreducible positive recurrent iff an invariant probability exists; uniqueness and Kac formula hold.
12. periodic two-cycle is irreducible but fails aperiodicity and TV convergence to equilibrium without parity restriction.
13. independent-coalescent implementation is generated from semantic product/diagonal laws, not OCR piecewise text.
14. Klenke (17.30) test uses factor 2; a two-point mutually singular pair catches a missing factor immediately.
15. TV convergence theorem targets total variation, not merely weak convergence.
16. IsQMatrix permits a general type with summable rows; countability appears only where construction needs it.
17. Q off-diagonal positivity vs diagonal negativity test catches lost $\neq$ OCR.
18. Lambda=0 uniformization yields $q=0$ and identity semigroup; no /0 shortcut.
19. uniformization with two different valid dominating rates gives the same semigroup by uniqueness.
20. Poisson Q-matrix recovers the existing Poisson transition law.
21. Klenke-style Feller definition is preserved; sup-norm strong continuity appears only through the audited equivalence theorem.
22. Feller operator is a contraction and its operator composition order matches kernel orientation.
23. Feller state-to-law weak continuity and $t \downarrow 0$ stochastic continuity reuse the measure-convergence package.
24. ordinary continuous-time Markov property alone does not imply strong Markov.
25. Klenke 21.27 cannot be marked completed unless the cited external proof was concretely accessed and audited; otherwise the blocker ledger says DEFERRED.
26. 21.27 conclusion is existence of an RCLL strong-Markov realization, not regularity of every realization.

15. Condensed Coverage / Ownership Ledger

Klenke / function	Status	Canonical directive
14.42 consistent transitions	BUILD likely	TransitionSystem first; forward-pair domain; Mathlib composition orientation.
14.43 Markov semigroup	BUILD likely	MarkovSemigroup property over Kernel family.
17.1 Markov property	BUILD likely	Kernel-free condExp event semantics; bounded-function theorem.
17.3/17.8 process/semigroup construction	BUILD/BRIDGE	Use StandardBorel KET; no process structure.
17.14 countable-time strong Markov	BUILD/BRIDGE	Use stopping-time infrastructure; general continuous time requires more.
17.16+ discrete matrix view	REUSE/BRIDGE	Kernel powers first; matrix view is secondary.
17.23-17.25 Q-matrix / bounded construction	BUILD	Summability predicate; uniformization; existence + uniqueness.
17.29-17.38 recurrence/Green	BUILD	Strict-positive hitting, Green, propagation.
17.44-17.52 invariant/positive recurrent	BUILD	Reuse Kernel.Invariant; excursion construction, uniqueness, Kac.
17.54 coupling	AUDIT-FIRST $\rightarrow$ BUILD likely	Generic probability coupling.
17.56 printed 17.30	SOURCE CORRECTION	Full TV norm requires factor 2.
18.1 periodicity	BUILD	Period + eventual positivity/cyclic structure.
18.5-18.6 successful/independent coalescent	BUILD	Path-law coupling; semantic pair kernel; OCR safeguard.
18.11-18.13 convergence	BUILD	Successful coupling $\rightarrow$ full TV convergence equivalences.

21.26 Feller	BUILD likely	Klenke pointwise C0 definition; Mathlib C0.
Feller $\leftrightarrow$ C0 strong continuity	EXTERNAL-PROOF-AUDIT $\rightarrow$ REUSE/BUILD/DEFER	Prefer Revuz-Yor III Prop.2.4 after exact source verification.
21.27 Feller RCLL strong Markov realization	EXTERNAL-PROOF-AUDIT $\rightarrow$ BUILD if accessible, else DEFER	Rogers-Williams III.7ff/8ff or Revuz-Yor III Thm.2.7; no autonomous proof if inaccessible.
MCMC / spectral rates	DEFER	Future application/finite-state extension.

16. Milestone Completion Gate

- Every BUILD is proved and every AUDIT-FIRST row is resolved against the actual pinned environment.
- Every external-proof item is either audited/formalized or explicitly DEFERRED because the proof text was inaccessible; there is no hidden dependency and no autonomous replacement proof.
- The source-correction table is implemented, especially kernel orientation, Klenke (17.30) factor 2, Example 18.6 semantic construction, and Q-matrix != OCR safeguard.
- The blocker/deferred ledger is fully updated with RESOLVED / STILL-BLOCKED / DEFERRED statuses and downstream impact.
- There are no sorry/admit/custom axioms; milestone theorems pass axiom/dependency inspection and semantic regression tests.
- No duplicate Mathlib Kernel/Invariant/Irreducible/Reversible/HittingTime/CondIndep/KET/C0 objects exist. Generic utilities are in generic owners.
- Imports are minimized and external namespaces do not leak into stable public API unless intentionally approved.
- Work submits exact source/revision/proof references sufficient for future maintainers to re-audit the package.

17. Source / Proof-Provenance Index for Work

- Primary coverage: Achim Klenke, Probability Theory: A Comprehensive Course, 3rd ed., Springer, 2020. Core Markov provenance: Ch.14.4, Ch.17-18; Feller/strong-Markov provenance: Def.21.26 and Thm.21.27.
- Klenke ref. [145]: D. Revuz and M. Yor, Continuous Martingales and Brownian Motion, Grundlehren der mathematischen Wissenschaften 293, 3rd ed., Springer, 1999. Klenke points to Ch.III, Thm.2.7 for the strong-Markov/Feller result. The commonly cited locator for Feller pointwise-to-strong-continuity equivalence is Ch.III, Prop.2.4; Work must verify the exact statement in the edition before formalization.
- Klenke ref. [147]: L.C.G. Rogers and D. Williams, Diffusions, Markov Processes, and Martingales, Vol.1: Foundations, Cambridge Mathematical Library, Cambridge UP, 2000 reprint of 2nd ed. Klenke points to Ch.III.7ff for RCLL Feller realization and Ch.III.8ff for strong Markov.
- Mathlib anchors identified during design reconnaissance (must be re-audited on the active pin):
Probability/Kernel/Composition/Comp; Kernel/Invariance; Kernel/Irreducible; Kernel/IonescuTulcea/Traj;
Probability/Process/{Filtration,Stopping,HittingTime}; MeasureTheory StandardBorel/C0/Portmanteau/ProbabilityMeasure infrastructure.
- For any external source not concretely accessible to Work, the prescribed response is DEFER + ledger entry, not an original proof attempt.